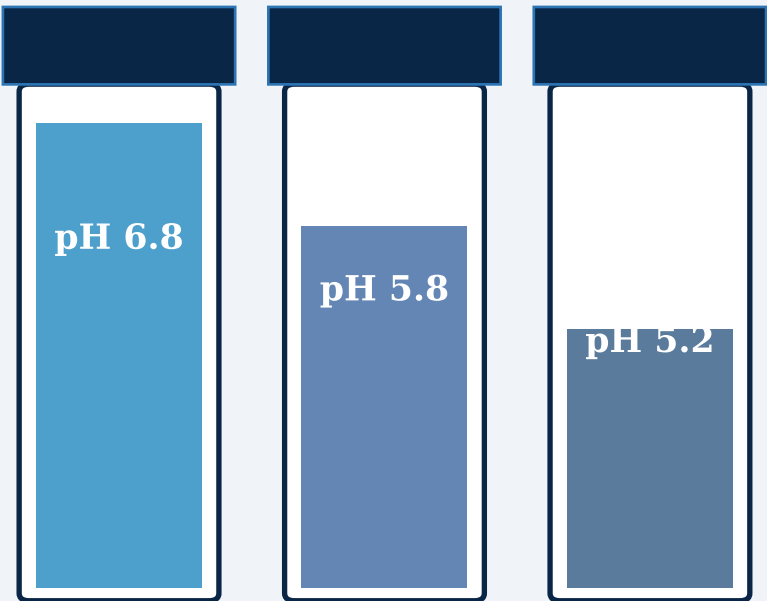


POSTMORTEM TISSUE pH ANALYSIS

DNA Methylation Clock Performance by Tissue Acidification Level

Postmortem Tissue

pH Degradation Timeline



Fresh
0-6h

Moderate
12-24h

Degraded
>48h

PMI (Postmortem Interval)

Tissue Acidification

pH 7.0 → 5.0

pH Category

Range

n

MAE

95% CI

R²

Status

Excellent Quality

pH >6.5

28

2.8 yr

2.1-3.6

0.93

OPTIMAL

Good Quality

pH 6.0-6.5

42

3.6 yr

3.0-4.3

0.89

GOOD

Moderate Quality

pH 5.5-6.0

26

5.1 yr

4.2-6.1

0.78

CAUTION

Poor Quality

pH <5.5

12

8.4 yr

6.7-10.3

0.52

NOT RECOMMENDED

FORENSIC NOTE

PMI-corrected Horvath clock

Total n = 108 postmortem samples

STATISTICAL ANALYSIS

ANOVA: $F(3,104) = 18.4, p < 0.001$ ***

Significant difference in MAE across tissue pH categories